DOMAIN REALIZATION PROFILE

Machine QA Evidence and Readiness Integration

Natural-Language Treatment Planning System (NL-TPS)

Source baseline: Controlled NL-TPS suite v0.1; local Monthly QA implementation snapshot reviewed 18 August 2026

Profile identifier: MQA-PKG-01

Profile version: 0.1 - Proposed baseline

Date: 18 August 2026

Status: Discussion Draft - Not for clinical use

Coverage: 12 realization requirements; 36 sub-requirements; 8 subassemblies; 12 principal risks; 8 trade-off decisions

STATUS AND SAFETY NOTICE

This profile defines how machine quality-assurance evidence may support NL-TPS readiness decisions. It does not approve a QA procedure, tolerance, commissioning reference, machine, treatment plan, or return to service. Qualified medical physicist ownership, approved institutional procedures, independent review, and existing treatment-delivery controls remain mandatory.

Contents

Document control

Field	Value
Document owner	NL-TPS Program; technical ownership assigned to the Qualified Medical Physicist and the Verification and Validation team
Source implementation	Local `Monthly QA` directory containing controlled-source extractors, workbooks, local retrieval corpora, normalized datasets, and the HUPTI QA Explorer prototype
Authority boundary	Approved departmental procedures, commissioned configurations, source QA records, and attributable QMP approvals remain authoritative
Change authority	Clinical governance, QMP technical authority, quality management, systems engineering, security/privacy, and independent verification
Review trigger	Procedure or tolerance revision; machine, detector, configuration, maintenance, software, extractor, schema, interface, or deployment change; incident; failed QA; source-integrity event; or scheduled review
Supersession	None

Executive summary

The reviewed Monthly QA implementation is a useful evidence and visualization prototype. It contains read-only extractors, longitudinal QA records, detector-specific provenance, commissioning candidates, Daily QA protocol-era separation, explicit missing-data behavior, local retrieval corpora, and a modern web explorer. Those capabilities can accelerate NL-TPS realization when they are placed behind the suite's existing governance, evidence, safety, security, state, audit, operations, and verification controls.

The implementation shall enter the NL-TPS as **MQA-PKG-01 Machine QA Evidence and Readiness Integration**, a domain package composed of existing controlled components. It shall not become a second planning authority, a replacement for source QA records, an automated QMP, or a retrieval-driven pass/fail engine. A signed, versioned readiness assertion may be consumed by the deterministic safety kernel only after the applicable QA evidence, approved rule, configuration, review, and return-to-service state have been reconciled.

NON-NEGOTIABLE CLINICAL BOUNDARY

The machine-QA subsystem may collect, normalize, display, compare, route, and preserve evidence. It may calculate a deterministic comparison using an approved computable rule. It shall not approve a procedure or tolerance, sign a QA result, waive a discrepancy, declare a machine safe, authorize return to service, approve a plan, or command treatment delivery. Retrieval and generative models shall not perform pass/fail or readiness adjudication.

1 Purpose and scope

This profile defines the controlled integration of Daily, Monthly, Annual, commissioning, maintenance-context, detector, and supporting QA evidence into the NL-TPS. It translates the reviewed prototype into requirements, interfaces, component allocations, data objects, workflows, risks, trade-off decisions, verification obligations, and implementation phases.

1.1 In scope

- Read-only acquisition from approved workbooks, reports, databases, DICOM objects, detector exports, and other commissioned source adapters.
- Canonical machine-QA observations with explicit units, configuration, protocol, tolerance, uncertainty, provenance, and lifecycle state.
- Longitudinal review, source drill-down, missing-evidence identification, protocol-era separation, and pre/post-maintenance comparison.
- QMP-controlled procedure, tolerance, reference, deviation, corrective-action, approval, expiry, and return-to-service workflows.
- Publication of a minimal signed machine-readiness assertion to the NL-TPS safety and proton-profile services.
- Natural-language query and draft-workflow support within the existing typed-intent and confirmation boundary.
- DICOM and external-TPS association only through commissioned, allowlisted interfaces.

1.2 Out of scope

- Autonomous approval of measurements, tolerances, procedures, references, deviations, or machine readiness.
- Substitution of statistical models, retrieval results, dashboards, or normalized data for authoritative source records.
- Silent conversion of a commissioning candidate, embedded expected value, historical baseline, or literature value into an approved clinical tolerance.
- Fabrication or imputation of missing measurements, dates, configurations, evidence files, or approvals.
- Direct treatment delivery, machine control, plan approval, QA waiver, or return-to-service action by AI or a service account.

2 Reviewed implementation baseline

2.1 Validated snapshot

The local prototype's built-in validators completed successfully for the following normalized snapshot. Counts describe the reviewed implementation and are not permanent clinical acceptance thresholds.

Domain	Validated snapshot	Interpretation
Monthly and archival	46 metric definitions; 6,579 observations; 2011–2026	Longitudinal GTR4/GTR5 records with explicit unavailable periods and no missing-month imputation
Detector evidence	5,204 observations; 33 evidence sources	Matrixx, MLIC, and Lynx measurements with source-level locators; two accepted prototype cases lack located raw Matrixx files
Annual QA	2,922 observations	2024–2025 records retain acquisition dates, reported replicates, detector configuration, and annual-report links
Daily QA	7,061 runs; 155,704 observations; 51 metrics	Legacy Access and Machine QA PDF protocol eras remain separate; a known GTR4 structured gap remains visible
Commissioning	21 reference sets; 304 points	Four comparison-ready candidate mappings; thirteen method-approval-required mappings; four inventory-only mappings; no approved reference set
Local retrieval	GTR4: 3,587 files and 13,057 chunks; GTR5: 4,468 files and 19,379 chunks	Local TF-IDF and latent semantic retrieval with SHA-256 manifests, source locators, privacy controls, and zero reported extraction errors

2.2 Reusable strengths

- Source-read-only and idempotent extraction intent.
- Explicit differentiation among measured values, embedded expected values, tolerances, and commissioning candidates.
- Source workbook, worksheet, cell, PDF page, raw-directory, raw-file, detector, date, and calculation-method locators.
- Preservation of repeated observations and configuration facets rather than silent averaging.
- Protocol-era separation and explicit structured-data gaps.
- Candidate, method-approval-required, and inventory-only commissioning states.
- Offline retrieval, source hashing, DICOM identifier redaction, excluded PLD patient columns, and metadata-only handling of proprietary binaries.
- Deterministic statistical functions and clear exploratory-use labeling for normal and Monte Carlo views.

2.3 Gaps requiring controlled redesign

- Application authentication, role authorization, electronic approval, immutable audit logging, deviation/CAPA workflow, and return-to-service gating are not implemented in the prototype.
- Large static JSON files are bundled with application code; Monthly/Annual data are passed to client components rather than queried through an authorized server boundary.

- Evidence records retain paths and filenames but are not directly bound to the RAG manifests' source hashes.
- Source paths, release expectations, record counts, transformation exceptions, and some reference mappings are encoded in scripts.
- Browser workbook loading accepts broad file extensions and performs only limited structural validation without a clinical schema signature, file-size limit, malware control, or resource sandbox.
- The health endpoint proves process liveness but not data integrity, evidence currency, configuration compatibility, or dependency readiness.
- The repository-wide lint command includes temporary browser-profile content; application-scoped source lint succeeds, but the release command is not currently reproducible without an ignore correction.

3 Authority and information model

3.1 Authority classes

Class	Examples	Permitted use	Prohibited use
Institutional procedure	Approved QA SOP, tolerance table, escalation and return-to-service policy	Generate approved computable rules within scope and effective dates	Infer policy from an unapproved template or dashboard
Authoritative QA record	Signed/final report, controlled workbook, approved database record, retained raw acquisition	Establish what was measured and reviewed	Replace with a retrieval chunk, screenshot, or recalculated display
Commissioning evidence	Approved configuration and reference measurement package	Support a scoped reference after QMP approval	Treat a candidate or inventory item as approved
Technical guidance	AAPM TG-224 and other approved technical sources	Support procedure design, uncertainty, and validation	Automatically create a local tolerance or release rule
Retrieval evidence	Indexed source text, metadata, and search results	Discovery, navigation, completeness review, and citation	Numerical calculation, pass/fail, readiness, signature, or approval
Derived analytics	Trends, statistics, distributions, simulations, and comparisons	Review support and anomaly screening	Override source status or establish clinical acceptability

3.2 Canonical objects

Object	Required responsibility
MQA Source Object	Immutable identity, source type, authoritative-system status, location, size, source hash, acquisition and modification times, sensitivity, retention, and access policy
MQA Import Run	Job identity, source scope, adapter and dependency versions, configuration, start/end time, counts, warnings, errors, reconciliation, reviewer, and output manifest
MQA Protocol Version	Controlled identifier, revision, owner, authority, scope, machine/configuration applicability, effective interval, measurement method, required evidence, escalation, review, and supersession
MQA Metric Definition	Stable metric identity, measurand, terminology, units, axes/configuration facets, calculation, rounding, display, and compatibility rules
MQA Tolerance Rule	Rule identity, comparator, expected or baseline source, limits, units, uncertainty treatment, scope, authority, approval, effective interval, exception policy, and tests
MQA Execution	Site, room, machine, beamline, configuration, QA program, protocol, scheduled/due state, acquisition time, operator, maintenance context, equipment, and source package
MQA Observation	Execution, metric, value, raw value, expected value, units, uncertainty reference, result, source locator, calculation lineage, review state, and anomaly links
MQA Deviation/CAPA	Affected observations and readiness scope, containment, investigation, impact, corrective/preventive action, effectiveness review, owners, approvals, and closure
MQA Readiness Assertion	Machine/configuration scope, normal/hold/conditional/released/expired state, applicable evidence and rules, unresolved conditions, QMP signature, effective interval, integrity value, and supersession

4 Architecture and interface allocation

4.1 Target data flow

The target architecture uses an asynchronous evidence-ingestion path and a minimal synchronous readiness path:

1. Approved source adapters read controlled archives without modifying them.
2. C-JOB-01 records an atomic import run and C-SEC-01 applies the approved parsing and privacy boundary.
3. C-TERM-01 and C-RULE-01 validate terminology, units, transformations, applicability, and approved comparison rules.
4. C-OBJ-01 stores immutable canonical versions; C-PROV-01 links source, transform, configuration, review, and downstream impact; C-AUD-01 records actions and decisions.
5. C-UX-01 presents observations, gaps, source evidence, trends, and review tasks to authorized users.

6. C-IAM-01, C-STATE-01, C-QMS-01, and C-INC-01 control review, approval, deviations, CAPA, and closure.
7. C-PBT-01 publishes the applicable signed readiness assertion to C-SAFE-01. The evidence explorer itself is not a synchronous treatment-planning dependency.

4.2 Existing interface profiles

Interface	Machine-QA profile	Required boundary behavior	Lead
IF-UI-01	Evidence and review workspace	Display scope, source authority, state, gaps, uncertainty, approval, and actions without color-only meaning	T-UX
IF-NLI-01	Typed machine-QA intent	Convert text or confirmed speech into query, comparison, draft-task, or source-navigation intent with deterministic validation	T-NLI
IF-IAM-01	QA role and approval	Enforce QMP, reviewer, operator, quality, informatics, and read-only privileges and separation of duties	T-SEC
IF-SAF-01	Readiness consumption	Accept only a current signed readiness assertion; fail closed on missing, mismatched, expired, or invalid state	T-SYS
IF-EVD-01	Procedure and tolerance rules	Return approved source, rule, applicability, effective date, conflict, and no-applicable-rule outcomes	T-EVD
IF-ACC-01	Quality evidence exchange	Exchange minimal evidence pointers, attestations, deficiencies, CAPA state, and readiness reporting without treatment authority	T-GOV
IF-DATA-01	QA objects and provenance	Store immutable versions, hashes, lineage, audit events, retention, and reconciliation	T-DATA
IF-CFG-01	Machine-QA release profile	Publish approved site, room, machine, mode, protocol, adapter, rule, monitor, and response configuration	T-GOV
IF-OPS-01	Import, health, alert, and recovery	Expose atomic job outcome, data age, integrity, dependency state, alerts, rollback, and recovery	T-OPS
IF-VV-01	Acceptance and commissioning evidence	Trace contracts to locked cases, expected results, anomalies, approval, and release evidence	T-VV
IF-DICOM-01 and IF-TPS-01	Commissioned association	Exchange only approved DICOM objects and bounded TPS transactions; retain UID, frame, machine, configuration, acknowledgment, and reconciliation	T-INT

4.3 DICOM and existing-TPS boundary

DICOM shall be used when a QA source is a valid DICOM object or when QA evidence must be associated with commissioned clinical objects. The subsystem shall not create synthetic DICOM objects merely to transport spreadsheets, reports, Access records, or proprietary detector formats. Those artifacts remain evidence objects in the canonical repository and may reference DICOM identifiers when a validated association exists.

Each external-TPS route shall have a site- and version-specific conformance analysis, transaction and object allowlist, identity and UID reconciliation, private-attribute policy, timeout and retry behavior, negative tests, known limitations, monitoring, and rollback. An undocumented private field or inferred equivalence shall not become a silent dependency.

5 Machine-QA realization requirements

The following requirements are subordinate realization requirements. They do not increase or replace the 119 high-level requirements in the controlled baseline.

ID	Requirement	Parent trace	Verify
MQA-001	The NL-TPS shall acquire machine-QA evidence only through approved, source-read-only adapters and shall record source scope, integrity, adapter version, outcome, warnings, and reconciliation for each atomic import.	DAT-002, DAT-009; SEC-004/005; OPS-002	I/T
MQA-002	The NL-TPS shall assign stable identity, version, ownership, lifecycle state, machine/configuration context, units, and source linkage to every machine-QA source, execution, metric, observation, rule, deviation, approval, and readiness assertion.	DAT-001/003/010; ACC-003/007	I/T
MQA-003	The NL-TPS shall evaluate a machine-QA observation only with an institution-approved, QMP-owned, versioned computable rule that is applicable to the active machine, configuration, protocol, method, and effective time.	EVD-002/005/006/007; ACC-006/008	I/T
MQA-004	The NL-TPS shall preserve bidirectional lineage from each displayed or applied result to the authoritative source object, exact locator, raw value, transformation, calculation, rule, software, configuration, user action, review, and downstream readiness effect.	DAT-002/008/009; ACC-003/007	I/T/A
MQA-005	The NL-TPS shall represent missing, incomplete, unreadable, unsupported, conflicting, raw-only, protocol-discontinuous, and unreviewed evidence explicitly and shall not impute or silently reconcile it.	SAF-004/010; DAT-007; EVD-005/008	T/A
MQA-006	The NL-TPS shall separate source-recorded results from deterministic recalculation and exploratory analytics and shall prevent a trend, fitted distribution, simulation, or retrieval result from changing pass/fail or readiness state.	SAF-008/009; REV-003/009; SEC-004	T/A/HFE
MQA-007	The NL-TPS shall enforce authenticated role authority, independent review, attributable QMP approval, effective time, and separation of duties for procedure, tolerance, exception, QA-result, and return-to-service decisions.	GOV-005/007; SAF-001/008/009; ACC-002/006	T/I
MQA-008	The NL-TPS shall support containment, discrepancy assessment, investigation, impact analysis, repeat/exclusion rationale, deviation, CAPA, effectiveness review, escalation, and closure without replacing the institutional quality process.	OPS-001/006; ACC-007; VAL-009	I/D/T
MQA-009	The NL-TPS shall publish and consume a minimal signed, versioned machine-readiness assertion and shall invalidate or hold affected scope after failed, missing, expired, superseded, changed, unreconciled, or postservice conditions until authorized resolution and return to service.	SAF-004/007/010; ACC-006/008; VAL-007/009	T/D

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ID	Requirement	Parent trace	Verify
MQA-010	The natural-language interface shall support machine-QA query, source navigation, comparison, completeness review, and draft-task preparation through inspectable typed intent while prohibiting AI approval, waiver, readiness, return-to-service, planning approval, or delivery authority.	NLI-002/005/006/008; SAF-001; SEC-004	T/HFE
MQA-011	The NL-TPS shall associate DICOM and external-TPS content only through commissioned object and transaction allowlists with identity, UID, frame, machine, configuration, unit, completeness, destination, and acknowledgment reconciliation.	DAT-004/005/007; REV-007; ACC-008	I/T
MQA-012	The machine-QA capability shall undergo threat analysis, privacy review, site acceptance, commissioning, reference and challenge testing, regression, shadow operation, recovery testing, monitoring approval, and governance release before its readiness output can affect clinical workflow.	SEC-001–008; OPS-001–007; VAL-001–009	I/T/D

6 Sub-requirements and component allocation

Each parent has a definition-and-schema child, an executable-behavior child, and an evidence-and-verification child. **U-STACK** means C-REQ-01, C-ARCH-01, C-CFG-01, C-PROV-01, C-AUD-01, and C-VV-01.

ID	Sub-requirement	Allocated realization
MQA-001-01	Define an allowlisted source-adapter contract covering source identity, access mode, schema, limits, privacy class, expected artifacts, hash policy, and error semantics.	U-STACK + C-SEC-01; C-JOB-01; C-TERM-01
MQA-001-02	Execute imports atomically in a resource-bounded environment; preserve source files; reject unsupported content; expose complete, partial, failed, canceled, or degraded outcomes.	C-JOB-01; C-SBX-01; C-SEC-01; C-RES-01
MQA-001-03	Retain import manifests, counts, hashes, warnings, errors, source-read verification, and reviewer reconciliation.	C-PROV-01; C-AUD-01; C-VV-01
MQA-002-01	Define canonical schemas and stable identifiers for every object listed in Section 3.2.	U-STACK + C-OBJ-01; C-TERM-01
MQA-002-02	Validate required identity, machine/configuration context, units, references, uniqueness, and lifecycle state before acceptance.	C-OBJ-01; C-TERM-01; C-SAFE-01; C-STATE-01
MQA-002-03	Test schema compatibility, migration, duplicate, orphan, wrong-machine, wrong-unit, and supersession behavior.	C-TEST-01; C-VV-01; C-PROV-01
MQA-003-01	Define each rule's authority, comparator, expected/baseline source, limits, units, uncertainty handling, scope, effective interval, conflict policy, and approval.	U-STACK + C-EVID-01; C-RULE-01; C-PBT-01
MQA-003-02	Resolve the highest applicable approved rule and return no-applicable-rule or conflict rather than silently selecting or blending.	C-RULE-01; C-SAFE-01; C-STATE-01
MQA-003-03	Verify boundary values, rounding, units, configuration, effective time, conflict, expiry, suspension, and rollback using locked cases.	C-TEST-01; C-VV-01; C-QMS-01
MQA-004-01	Define source, locator, raw value, transformation, calculation, rule, software, configuration, actor, review, and downstream dependency links.	U-STACK + C-OBJ-01; C-PROV-01
MQA-004-02	Prevent publication when a mandatory provenance edge or integrity value is missing or inconsistent.	C-PROV-01; C-SAFE-01; C-STATE-01
MQA-004-03	Demonstrate source-to-display and display-to-source reconstruction for representative and challenged records.	C-VV-01; C-TEST-01; C-AUD-01
MQA-005-01	Define explicit evidence states and protocol-era compatibility rules without a generic acceptable default.	U-STACK + C-EVID-01; C-RULE-01; C-STATE-01

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ID	Sub-requirement	Allocated realization
MQA-005-02	Display gaps and incompatibilities and block affected promotion when approved policy requires the evidence.	C-UX-01; C-SAFE-01; C-PBT-01; C-MON-01
MQA-005-03	Test missing months, blank formulas, absent raw files, unreadable files, protocol gaps, conflicting dates, and unsupported mappings.	C-TEST-01; C-VV-01
MQA-006-01	Define distinct fields and visual states for source result, deterministic comparison, trend statistic, fitted model, simulation, and retrieval evidence.	U-STACK + C-OBJ-01; C-UX-01; C-HFE-01
MQA-006-02	Permit only the approved deterministic rule to affect evaluation state; isolate analytics and retrieval from the readiness transaction.	C-RULE-01; C-SAFE-01; C-STATE-01; C-SBX-01
MQA-006-03	Validate normality caveats, small samples, outliers, extreme scales, alert comprehension, and attempts to promote exploratory output.	C-HFE-01; C-TEST-01; C-VV-01
MQA-007-01	Define role, purpose, independence, signature, scope, effective time, credential, competency, and expiration for each controlled action.	U-STACK + C-IAM-01; C-HFE-01; C-QMS-01
MQA-007-02	Enforce role and state authorization; prohibit AI and service-account signatures and incompatible self-approval.	C-IAM-01; C-SAFE-01; C-STATE-01
MQA-007-03	Retain attributable decisions and test unauthorized, expired, wrong-purpose, shared-account, and separation-of-duties cases.	C-AUD-01; C-VV-01; C-TEST-01
MQA-008-01	Define deviation, containment, investigation, impact, action, effectiveness, escalation, closure, and reopening states and owners.	U-STACK + C-QMS-01; C-INC-01; C-STATE-01
MQA-008-02	Link every discrepancy to affected evidence and readiness scope and prevent silent deletion, exclusion, repetition, or closure.	C-INC-01; C-PROV-01; C-SAFE-01; C-MON-01
MQA-008-03	Verify escalation, overdue action, repeat rationale, effectiveness, reopening, audit, and authorized exception paths.	C-VV-01; C-TEST-01; C-AUD-01
MQA-009-01	Define readiness scope, state, prerequisites, unresolved conditions, signature, integrity, effective interval, expiry, and invalidation triggers.	U-STACK + C-PBT-01; C-QA-01; C-STATE-01
MQA-009-02	Publish and consume readiness atomically; fail closed on absent, mismatched, expired, invalid, postservice, or unreconciled state.	C-PBT-01; C-SAFE-01; C-JOB-01; C-MON-01
MQA-009-03	Test maintenance, repair, software change, failed QA, expiry, rollback, reapproval, degraded operation, and return-to-service scenarios.	C-VV-01; C-TEST-01; C-RES-01
MQA-010-01	Define allowlisted query, comparison, completeness, source-navigation, and draft-task intents and prohibited approval or delivery intents.	U-STACK + C-INTENT-01; C-WF-01; C-UX-01
MQA-010-02	Validate machine, period, metric, units, configuration, evidence scope, action, and authority; present read-back and require confirmation for state-changing draft actions.	C-INTENT-01; C-SAFE-01; C-IAM-01; C-UX-01
MQA-010-03	Test ambiguity, prompt injection, embedded instructions, wrong room, wrong period, prohibited action, correction, cancellation, and audit reconstruction.	C-TEST-01; C-HFE-01; C-AUD-01; C-SEC-01
MQA-011-01	Define the approved DICOM and TPS object/transaction profile, identities, roles, UIDs, frames, units, configurations, private-field policy, and acknowledgments.	U-STACK + C-DICOM-01; C-DICOM-02; C-TPS-01
MQA-011-02	Reject incomplete, wrong-context, orphaned, partial, unsupported, or unreconciled transfers without substituting a file-based shortcut.	C-DICOM-02; C-SAFE-01; C-JOB-01
MQA-011-03	Execute conformance, round-trip, duplicate, timeout, retry, partial-transfer, private-attribute, destination, and reconciliation tests for each released route.	C-VV-01; C-TEST-01; C-TPS-01

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ID	Sub-requirement	Allocated realization
MQA-012-01	Define the assurance plan, hazards, critical tasks, reference cases, challenge cases, monitors, thresholds, recovery, and release evidence.	U-STACK + C-GOV-01; C-QMS-01; C-MON-01
MQA-012-02	Progress through retrospective, commissioning, shadow, and limited controlled stages without permitting readiness influence before release approval.	C-SBX-01; C-VV-01; C-SAFE-01; C-CFG-01
MQA-012-03	Block or suspend service for unresolved critical anomalies, unacceptable residual risk, failed thresholds, unverified controls, integrity failures, or unreconciled change.	C-SAFE-01; C-MON-01; C-INC-01; C-RES-01

7 Package subassemblies

MQA-PKG-01 is a deployment profile and coordinated realization package, not a new independent authority. Its subassemblies allocate work to the existing component baseline.

ID	Subassembly	Responsibility	Primary components
MQA-A01	Controlled Source Adapter Set	Read-only, allowlisted, resource-bounded source discovery, extraction, hashing, error reporting, and reconciliation	C-ARCH-01; C-SEC-01; C-JOB-01; C-SBX-01
MQA-A02	Canonical QA Object Store	Versioned source, execution, observation, evidence, approval, deviation, and readiness objects with units and lineage	C-OBJ-01; C-TERM-01; C-PROV-01; C-AUD-01
MQA-A03	Protocol and Tolerance Registry	Approved QMP-owned procedure, applicability, baseline, tolerance, uncertainty, effective-date, and conflict rules	C-EVID-01; C-RULE-01; C-CFG-01; C-PBT-01
MQA-A04	QA Review and Quality Workflow	Review, approval, signature, deviation, CAPA, escalation, expiry, invalidation, and closure	C-IAM-01; C-STATE-01; C-QMS-01; C-INC-01
MQA-A05	Evidence and Analytics Workspace	Configuration-aware observations, trends, gaps, sources, deterministic results, and clearly separated exploratory analytics	C-UX-01; C-MON-01; C-HFE-01
MQA-A06	Machine Readiness Publisher	Minimal signed readiness assertion, invalidation, hold, expiry, and return-to-service integration	C-PBT-01; C-QA-01; C-SAFE-01; C-VV-01
MQA-A07	Natural-Language QA Interaction	Typed query, comparison, navigation, and draft-task intent with read-back, confirmation, and prohibited actions	C-INTENT-01; C-WF-01; C-UX-01; C-SAFE-01
MQA-A08	External QA Association Profile	Commissioned DICOM/TPS linkage and minimal quality-evidence exchange	C-DICOM-01; C-DICOM-02; C-TPS-01; C-ACC-02

8 Workflow and readiness state

8.1 Evidence workflow

State	Entry condition	Permitted transition
Discovered	Source identified under an approved adapter scope	Import pending or excluded with rationale
Imported	Atomic import completed with manifest and source integrity	Validation pending; never equivalent to approved
Validated	Schema, identity, units, references, transforms, and rule applicability pass	Human review pending
Under review	Authorized reviewer examines source, result, gaps, repeats, and context	Approved, discrepancy, rejected, or more information required
Approved evidence	Required attributable review is complete	May contribute to readiness within scope and effective time
Discrepancy/hold	Failure, missing evidence, conflict, invalidation, service event, or unresolved condition exists	Investigation, correction, retest, exception, or retirement
CAPA	Controlled quality workflow is active	Effectiveness review and authorized closure
Superseded	A newer approved version or retirement decision exists	Read-only retention; never returned silently to active use

8.2 Readiness states

Readiness is a separate object from evidence review. **Unknown** means the applicable state cannot be established. **Hold** means an approved rule requires clinical-use restriction. **Conditional** means a specifically authorized limited scope and condition exist. **Released** means all applicable prerequisites and QMP authorization are current. **Expired** means the effective interval ended. Every state shall identify exact machine/configuration scope and shall be consumed through C-SAFE-01 rather than inferred from dashboard color or record count.

9 Natural-language operating profile

Intent class	Examples	Required response
Query and navigate	“Show GTR5 range constancy for July”; “open the source for this point”	Resolve typed room, period, metric, configuration, and source; display provenance and evidence state

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Intent class	Examples	Required response
Compare	"Compare the approved pre-PM and post-PM output records"	Identify compatibility, rules, gaps, uncertainty, and exact source versions; abstain on unapproved equivalence
Completeness review	"Find missing raw detector evidence for this package"	Return explicit required, present, missing, excluded, and unresolved evidence with approved checklist source
Draft workflow	"Prepare a draft deviation for review"	Present read-back, affected scope, source links, proposed fields, and confirmation; create only a draft
Prohibited authority	"Approve this tolerance"; "return the room to service"; "waive this failure"	Refuse the action, identify the authorized role and controlled workflow, and record the attempt as required

Retrieved documents, workbooks, DICOM metadata, and reports are untrusted content under SEC-004. Instructions embedded in those artifacts shall never change system policy, authorize tools, alter the typed intent, or override user and institutional authority.

10 Source migration disposition

Prototype capability	Disposition	Controlled implementation decision
Read-only extractors	Reuse after validation	Convert paths, schemas, transformations, privacy rules, limits, and expected artifacts to approved configuration; sign and test adapter releases
Normalized TypeScript/JSON types	Refactor	Promote to versioned canonical schemas with runtime validation, migration, referential integrity, effective time, approvals, and hashes
Trend and provenance views	Reuse/refactor	Preserve interaction patterns; query authorized server APIs; distinguish evidence and analytics; perform human-factors validation
Static JSON deployment	Replace	Use immutable versioned storage and paged, filtered, authorized server queries; publish a signed release manifest
Browser workbook loader	Sandbox only	Restrict to training/commissioning unless separately approved; validate signature/schema/type/size; scan and isolate content
Local RAG corpus	Reuse for discovery	Preserve local retrieval, source hashes, redaction, and citations; prohibit numerical and readiness authority; sign retrieval artifacts before controlled use

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Prototype capability	Disposition	Controlled implementation decision
Commissioning candidates	Preserve as candidates	Require QMP approval of version, method, units, interpolation, uncertainty, tolerance, configuration, and effective dates before use
Normal and Monte Carlo views	Exploratory only	No connection to pass/fail or readiness; validate labeling, assumptions, small-sample behavior, and user comprehension
Simple health endpoint	Replace/extend	Report application, data revision, integrity, evidence age, dependencies, configuration compatibility, degraded state, and last successful reconciliation

11 Risk and mitigation register

Risk	Hazard scenario	Initial	Required mitigation and evidence
MQA-R01	Wrong tolerance, expected value, baseline, comparator, or effective date creates an incorrect evaluation.	Critical	QMP-approved computable rule; version/effective scope; unit and boundary tests; dual review; fail closed on conflict or no rule
MQA-R02	Normalized result no longer matches its source or transformation.	High	Full-file hash; exact locator; raw/derived separation; adapter and dependency version; transformation trace; source-to-display reconstruction
MQA-R03	Missing or incomplete evidence appears acceptable.	Critical	Explicit evidence states; required-artifact profile; no imputation; completeness gate; negative tests; visible unresolved condition
MQA-R04	Candidate commissioning data are treated as approved references.	Critical	Candidate/method-pending/inventory/approved lifecycle; separate fields and visuals; QMP signature; effective configuration; promotion tests
MQA-R05	Retrieval or model output performs pass/fail or readiness adjudication.	Critical	Typed data and deterministic rules only; tool separation; prompt-injection tests; prohibited intent; audit; independent safety-kernel tests
MQA-R06	Shared calculation logic defeats independent verification.	High	Independence analysis; source-result preservation; separately derived locked cases; common-cause documentation; QMP review
MQA-R07	Stale QA remains associated with an active machine/configuration after maintenance or change.	Critical	Dependency graph; service/change triggers; atomic invalidation; hold state; expiry; reconciliation; QMP return-to-service signature
MQA-R08	Untrusted document or oversized workbook compromises the application or changes behavior.	High	Sandboxed parsing; size/time/memory limits; allowlisted formats and schemas; malware controls; SEC-004 instruction isolation
MQA-R09	Internal paths, workforce identifiers, or sensitive metadata cross an unapproved boundary.	High	Minimum necessary views; RBAC; local evidence handling; redaction/pseudonym policy; encrypted transport; export review; audit
MQA-R10	Statistical model assumptions or scale effects mislead reviewers.	High	Exploratory labeling; raw-value preservation; focus/full-scale controls; sample and assumption disclosure; HFE validation; no readiness linkage
MQA-R11	Transfer associates QA evidence with the wrong machine, plan, frame, or destination.	Critical	Commissioned DICOM/TPS route; identity/UID/frame/configuration validation; acknowledgments; round-trip and wrong-context tests

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Risk	Hazard scenario	Initial	Required mitigation and evidence
MQA-R12	Service failure, partial import, rollback, or restore presents stale or partially reconciled readiness.	Critical	Atomic jobs; signed release manifest; explicit degraded mode; staged recovery; restore and reconciliation tests; fail-closed readiness consumption

Residual risk shall be assessed after mitigation evidence is available. No row in this register constitutes risk acceptance.

12 Trade-off analysis

Decision	Option A	Option B	Selected approach and rationale
Prototype integration	Merge the dashboard directly into the clinical runtime	Rebuild every capability from zero	Reuse validated extraction and presentation patterns behind controlled services and schemas. This retains evidence value without importing prototype authority or deployment assumptions.
Runtime dependency	Make the full evidence explorer synchronous with planning	Keep QA wholly disconnected from planning	Publish a minimal signed readiness assertion synchronously; keep evidence ingestion and exploration asynchronous.
Evaluation method	Let retrieval or AI infer pass/fail	Use workbook formulas as the permanent authority	Use versioned, QMP-approved deterministic rules while retaining source-recorded results and formula provenance.
Data delivery	Ship complete static JSON to each client	Query authoritative objects through server APIs	Use paged, filtered, authorized server queries and immutable releases to reduce exposure, payload, and stale-data risk.
Workbook replacement	Permit unrestricted browser upload in clinical mode	Prohibit all ad hoc workbook review	Confine replacement workbooks to segregated training/commissioning unless a validated controlled-ingestion workflow is approved.
Interface architecture	Create a new top-level interface family immediately	Reuse existing interfaces without a domain profile	Reuse IF-ACC, IF-EVD, IF-DATA, IF-CFG, IF-OPS, IF-VV, IF-SAF, IF-DICOM, and IF-TPS with this explicit machine-QA profile.
DICOM use	Convert every QA artifact into DICOM	Avoid DICOM for QA entirely	Use DICOM only for valid objects and commissioned associations; retain non-DICOM artifacts as governed evidence with referenced UIDs where applicable.
Analytics	Remove statistical and simulation tools	Let analytics drive automated action	Retain clearly separated exploratory analytics for review and research; prohibit readiness authority and verify user comprehension.

13 Verification and acceptance

13.1 Locked reference and challenge set

The initial controlled test library shall include, at minimum:

- representative GTR4 and GTR5 Daily, Monthly, Annual, and commissioning records;
- pre-maintenance, post-maintenance, routine, missing-month, raw-only, and protocol-

transition examples;

- output, flatness, symmetry, range, spot size, spot position, mechanical, imaging, and detector-specific configurations;
- exact boundary, just-inside, just-outside, unit-conversion, rounding, and no-applicable-rule cases;
- missing raw evidence, blank template formulas, duplicate observations, repeated measurements, inconsistent dates, wrong room, wrong configuration, and corrupted or unsupported files;
- candidate, method-pending, inventory-only, approved, expired, suspended, and superseded reference states;
- unauthorized approval, incompatible self-review, prompt injection, embedded document instructions, and attempted AI return-to-service actions;
- DICOM wrong-patient, wrong-study, wrong-frame, orphaned reference, duplicate, partial, timeout, and destination-reconciliation cases;
- maintenance, repair, software upgrade, rollback, data restore, degraded mode, invalidation, retest, and return-to-service scenarios.

13.2 Release gates

MQA-PKG-01 shall not influence clinical readiness until:

1. The source inventory, authority, privacy, retention, ownership, and integrity baseline are approved.
2. Canonical schemas, terminology, units, transformations, procedures, tolerances, and effective-date rules are controlled and traceable.
3. Every released adapter is source-read-only, resource-bounded, signed, independently reviewed, and tested against locked evidence.
4. Authentication, role authorization, electronic signatures, separation of duties, audit integrity, deviation/CAPA, and return-to-service workflows are verified.
5. Retrieval and analytics are demonstrably isolated from pass/fail, readiness, approval, and treatment authority.
6. Readiness invalidation, expiry, hold, degraded operation, rollback, backup, restore, and reconciliation are verified end to end.
7. Each DICOM/TPS route passes conformance, object, transaction, identity, UID, frame, unit, private-field, round-trip, and destination tests.
8. Representative users complete formative and summative human-factors testing for source authority, gaps, warnings, protocol eras, candidate references, confirmations, and recovery.
9. Retrospective and shadow-mode results meet approved accuracy, completeness, false-positive, false-negative, availability, latency, security, and user-burden thresholds.

10. The QMP, clinical governance, quality, security/privacy, informatics, operations, and independent V&V authorities approve residual risk and the released scope.

14 Controlled implementation sequence

1. **Baseline and freeze.** Inventory a representative evidence package, record hashes and authority, confirm August 2026 and later source status directly from the controlled share, and preserve a locked test snapshot.
2. **Control the information model.** Approve canonical objects, terminology, units, protocol and tolerance schemas, transformations, lifecycle states, and source-authority rules.
3. **Harden ingestion.** Convert extractors into signed configuration-driven jobs with atomic output, resource limits, privacy controls, integrity linkage, and reconciliation reports.
4. **Deploy read-only exploration.** Reuse the strongest chart, configuration, gap, and source-drill-down patterns through authenticated server APIs in a segregated mode.
5. **Add human governance.** Implement role-based review, signatures, procedure/tolerance approval, deviations, CAPA, expiry, and audit reconstruction.
6. **Publish readiness.** Implement the minimal signed readiness assertion, deterministic safety-kernel consumption, invalidation, hold, degraded operation, and QMP return to service.
7. **Add natural-language support.** Release allowlisted query and draft-workflow intents after ambiguity, prohibited-action, prompt-injection, read-back, confirmation, cancellation, and HFE testing.
8. **Commission external routes.** Add DICOM and TPS associations one site, version, machine, object, and transaction profile at a time.
9. **Validate and release.** Complete reference/challenge tests, shadow operation, recovery, security, performance, human-factors, site acceptance, residual-risk approval, and governed release.

15 Definition of done

The profile is realized only when every MQA requirement and sub-requirement has an approved owner, design allocation, interface contract, hazard and risk trace, implementation, test case, expected result, executed result, anomaly disposition, monitoring signal, rollback method, and release decision. The operational system must be able to reconstruct why an observation was displayed, which source and rule were used, who reviewed it, what readiness scope it affected, and why the resulting state was accepted or blocked.

Bottom line. The prototype is a valuable evidence and interface foundation; clinical use requires a governed, typed, traceable, QMP-owned readiness service. Accountability remains explicit: clinical physics owns technical meaning and return to service; quality owns process/CAPA; systems engineering owns deterministic state/safety; data and security own integrity/privacy; integration owns commissioned DICOM/TPS boundaries; independent V&V owns release evidence.